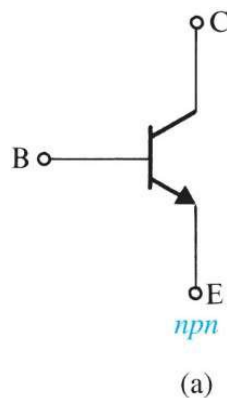


Outline

- › Physical structure
- › Current-voltage relationships
 - Regions (modes) of operation
- › BJT circuits

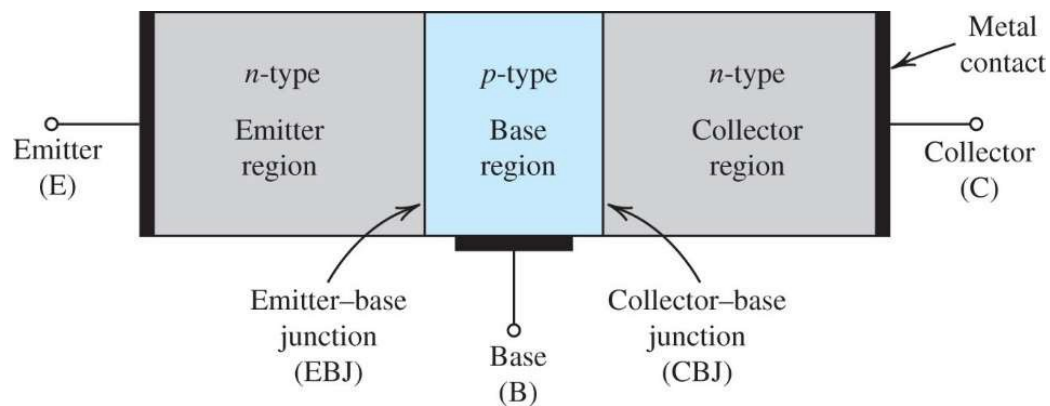
Introduction

- › BJT is a three-terminal device: **Base, Collector, Emitter**
- › Transistor (BJT) invented in 1948 at Bell labs
- › A new era of semiconductor devices and circuits
- › BJT not as common as MOSFETs in modern electronics
- › BJTs are used primarily in discrete-component (but also integrated) circuits



Device structure and physics

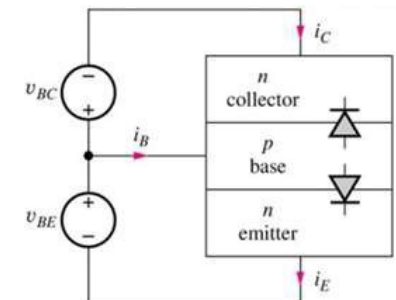
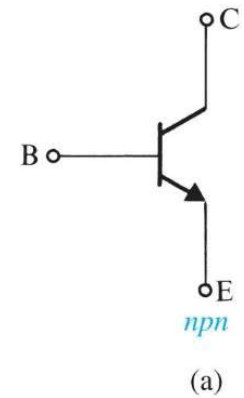
- › BJT consists of three semiconductor regions:
 - Emitter region: n-type
 - Base region: p-type
 - Collector region: n-type
- › Type described is npn. pnp types exist as well.



A simplified structure of the npn transistor

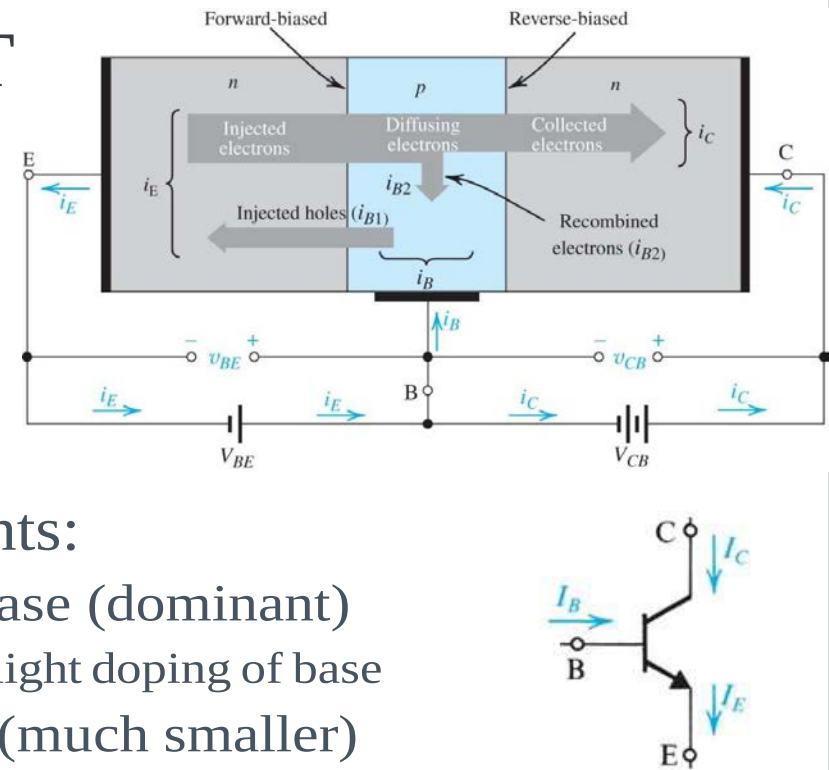
Modes of operation

- › Two pn junctions:
 - Base to emitter
 - Base to collector
- › Three modes of operation:
 - **Active** mode – used primarily for **amplification**
 - **Saturation** mode – used for primary for **switching**
 - **Cutoff** mode – device is off
- › Operating mode depends on biasing
- › Two external voltage sources are required for biasing
 - A source at the base (to bias B-E junction)
 - A source at the collector (to bias B-C junction)



Current flow in an npn BJT

- › Current flows when B-E junction is forward-biased
- › Recall that direction of current flow is opposite to the direction of electron flow
- › This current (i_E) has two components:
 - Electrons injected from emitter into base (dominant)
 - › is achieved with heavy doping of emitter, light doping of base
 - Holes injected from base into emitter (much smaller)
- › The electrons coming from the emitter will diffuse through the base and reach the collector
- › Thus, current flows out of the emitter and into the collector



Collector current in the forward active region

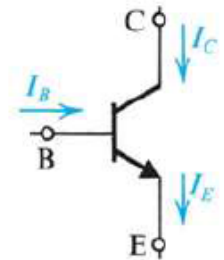
- › The current across a forward-biased pn junction is proportional to $e^{\frac{v_{BE}}{V_T}}$

- where v_{BE} is the forward voltage across the base-emitter junction
- and V_T =thermal voltage = $kT/q = 0.025$ V at room temperature

$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

- where I_S =saturation current; typically 10^{-18} A < I_S < 10^{-9} A. I_S is proportional to the area of the B-E junction and inversely proportional to the width, W , of the base region.

Note that i_C is independent of v_{BC} in the forward active region

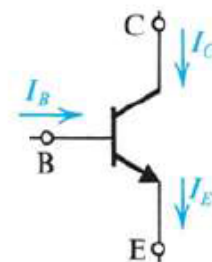


Base current in the forward active region

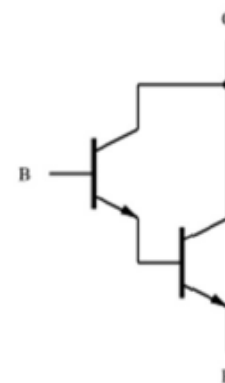
- › The base current in the **active region** can be expressed as:

$$i_B = \frac{i_C}{\beta}$$

$$i_B = \frac{I_S}{\beta} e^{\frac{v_{BE}}{V_T}}$$



- › β is known as current gain; typically, $20 < \beta < 500$
- › β can be higher for some special devices
- › A Darlington pair of BJTs increases β



Emitter current in the forward active region

- › Current that enters a transistor must leave it.

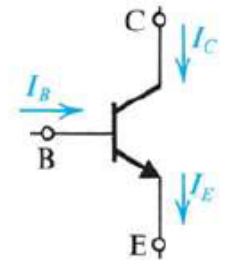
$$i_E = i_C + i_B$$

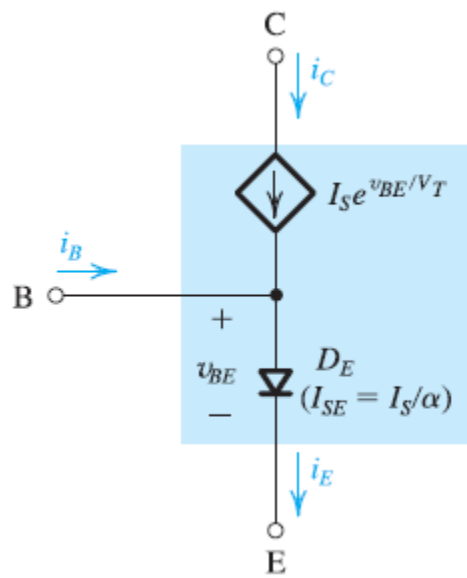
$$i_E = \frac{\beta + 1}{\beta} i_C = \frac{i_C}{\alpha}$$

$$\alpha = \frac{\beta}{\beta + 1} \text{ or } \beta = \frac{\alpha}{\alpha - 1}$$

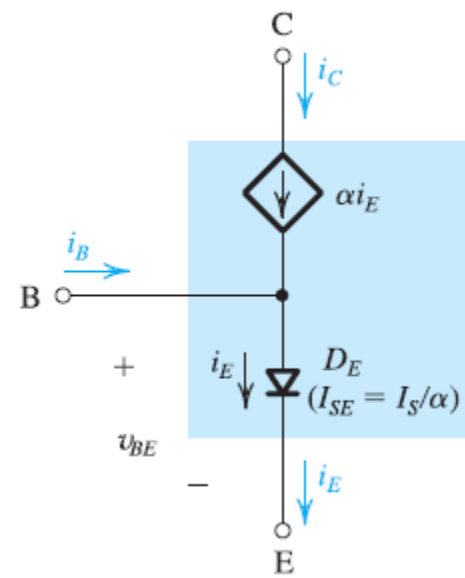
$$i_E = \frac{\beta + 1}{\beta} I_S e^{\frac{v_{BE}}{V_T}} = \frac{I_S}{\alpha} e^{\frac{v_{BE}}{V_T}}$$

- › parameter α : $0.95 < \alpha < 1$

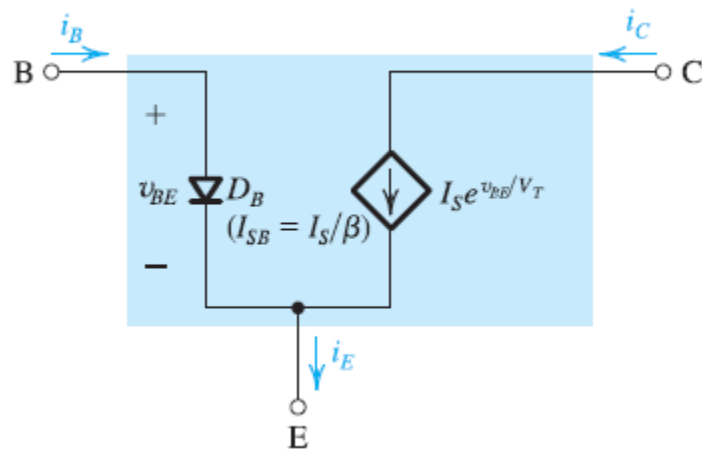




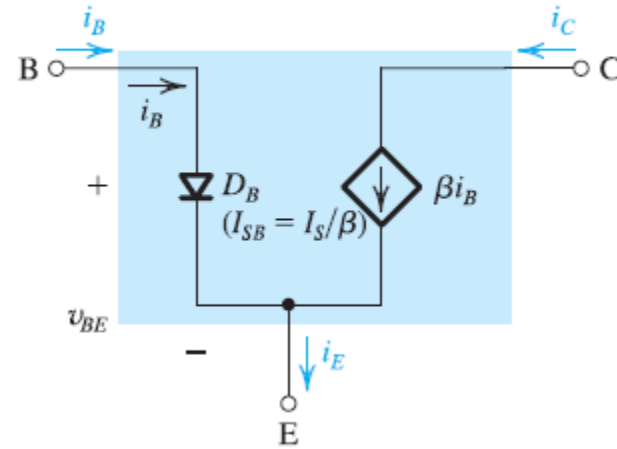
(a)



(b)



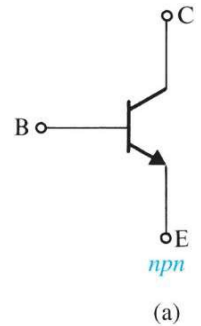
(c)



(d)

Large-signal equivalent-circuit models of the *n*p*n* BJT operating in the forward active mode.

BJT (npn) modes of operation



Base-Emitter Junction	Base-Collector Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)